Hybrid Quantum-Classical Machine Learning for High-Dimensional Biomedical Disease Detection: A Rigorous Methodology and Benchmarking Study

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***Abstract*—*High-dimensional biomedical datasets pose fundamental statistical challenges to classical diagnostic architectures, predominantly manifested as the curse of dimensionality, non-linear geometric entanglement, and sample-to-feature ratio vulnerabilities. In this study, we present a collaborative, leak-free Hybrid Quantum-Classical Machine Learning (HQML) framework tailored for contemporary Noisy Intermediate-Scale Quantum (NISQ) processors (4–8 physical qubits). We benchmark our architecture across two gold-standard clinical cohorts: the Wisconsin Diagnostic Breast Cancer (WDBC, N=569, D=30) and the UCI Cleveland Heart Disease (N=303, D=13) repositories. Our contributions comprise: (i) deterministic, leak-free data ingestion and preprocessing pipelines enforcing stratified 5-fold cross-validation; (ii) dual-path dimensionality reduction matching hardware qubit constraints via Principal Component Analysis (scaled to [−π, π] for ZZ-Feature Maps) and Mutual Information ranking (scaled to [0, 2π] for parameterized rotation gates); (iii) a comprehensive suite of four classical baselines—Linear/RBF Support Vector Machines (SVM), a 3-layer regularized PyTorch Multi-Layer Perceptron (MLP), Random Forests, and XGBoost; and (iv) mathematical formulations for quantum Hilbert space feature mapping, variational quantum circuits (VQC), and quantum kernel estimation (QKE). Classical benchmarks establish a robust validation standard, achieving 97.72 ± 1.1% accuracy on WDBC and 84.82 ± 2.1% on Heart Disease, providing the definitive empirical baseline for evaluating quantum advantage in biomedical diagnostics.*

***Index Terms*—**Hybrid Quantum-Classical Machine Learning, Wisconsin Diagnostic Breast Cancer, UCI Heart Disease, Quantum Kernel Estimation, Variational Quantum Classifier, Leak-Free Preprocessing, Explainable AI.

I. INTRODUCTION

Early detection of pathophysiological conditions—including invasive ductal breast carcinomas and coronary artery occlusions—represents the cornerstone of modern preventative medicine. In clinical pathology and diagnostic cardiology, patient cohorts are characterized by dense, multi-modal feature vectors derived from fine-needle aspirates (FNA) and non-invasive cardiovascular stress tests. Despite recent advances in deep learning, classical statistical classifiers operating in moderate-sized patient cohorts frequently encounter severe constraints: (1) *The Curse of Dimensionality*: as feature space dimensionality D expands relative to sample size N, data points become exponentially equidistant in Euclidean space, deteriorating the margin-separation efficacy of standard hyperplanes; (2) *Data Leakage in Biomedical Pipelines*: pre-scaling or selecting features across pooled datasets prior to cross-validation splits yields spuriously inflated accuracy; and (3) *NISQ Hardware Qubit Bottlenecks*: contemporary physical quantum devices are restricted to 4–8 active qubits due to two-qubit gate error rates and decoherence.

To address these challenges, this paper formalizes the end-to-end data engineering, dimensionality reduction, and classical baseline framework developed by Member A, designed to seamlessly interface with Member B's quantum circuit architectures and Member C's clinical explainability audit.

II. MATHEMATICAL FORMULATION OF HYBRID QUANTUM-CLASSICAL SYSTEMS

A. Quantum Hilbert Space Mapping & ZZ-Feature Maps

Let a patient feature vector be denoted as $\mathbf{x} = [x_1, x_2, ..., x_k]^T \in \blacksquare^k$, where k matches the physical qubit register size. A quantum feature map defines an embedding into a 2^k-dimensional complex Hilbert space $\blacksquare$:

$$\Phi: \blacksquare^k \rightarrow \blacksquare, \; x \blacksquare |\Phi(x)\blacksquare = U_\Phi(x)|0\blacksquare^{\otimes k}$$

For Member B's second-order Pauli *ZZ*-Feature Map, the unitary operator acting on the ground state is defined as:

$$U_{\Phi}(\mathbf{x}) = \exp\left(i \sum_{j} \phi_j(\mathbf{x}) Z_j + i \sum_{j<l} \phi_{jl}(\mathbf{x}) Z_j Z_l \right) H^{\otimes k}$$

where $\phi_j(\mathbf{x}) = x_j$ and $\phi_{jl}(\mathbf{x}) = (\pi - x_j)(\pi - x_l)$. When inputs are scaled to $[-\pi, \pi]$ via our PCAReducer, this transformation generates non-linear correlations with high quantum circuit expressibility.

B. Quantum Kernel Estimation (QKE)

For dual-form Support Vector Machines, quantum fidelity defines the Mercer kernel:

$$K(\mathbf{x}_i, \mathbf{x}_j) = |\langle \Phi(\mathbf{x}_i) | \Phi(\mathbf{x}_j) \rangle|^2 = |\langle 0^{\otimes k} | U_{\Phi}^{\dagger}(\mathbf{x}_j) U_{\Phi}(\mathbf{x}_i) | 0^{\otimes k} \rangle|^2$$

Solving the dual convex quadratic program optimizes support vector coefficients α_i subject to $0 \le \alpha_i \le C$ and $\sum \alpha_i y_i = 0$.

C. Variational Quantum Classifier (VQC)

In a VQC, a parameterized ansatz $U(\theta)$ processes the encoded state via layers of single-qubit rotations and entangling CNOT rings: $|\psi(\mathbf{x}, \theta)\rangle = U(\theta)|\Phi(\mathbf{x})\rangle$. The expectation value of Pauli-Z on the readout qubit yields the risk score $\hat{y}(\mathbf{x}) = \text{sign}(\langle \psi | Z_1 | \psi \rangle - b)$.

III. RIGOROUS LEAK-FREE DATA ENGINEERING METHODOLOGY

A. Wisconsin Diagnostic Breast Cancer (WDBC) Pipeline

The WDBC dataset comprises N=569 patient biopsies (212 malignant, 357 benign) with 30 continuous nuclear morphological features. Crucially, feature scaling parameters (mean μ and variance σ^2) are computed exclusively on training folds and subsequently applied to validation partitions, preventing data leakage. Verification confirms $\mu = 0.0 \pm 10^{-7}$ and $\sigma = 1.0 \pm 10^{-7}$ on training folds.

B. UCI Cleveland Heart Disease Pipeline

The Cleveland cohort incorporates N=303 records. Target diagnosis is binarized (num > 0 → 1 [139 disease cases]; num = 0 → 0 [164 healthy]). Missing attributes in fluoroscopy major vessels (*ca*, 4 missing) and thallium scintigraphy (*thal*, 2 missing) are imputed with training-fold medians. Categorical variables are one-hot encoded, and numerical predictors are standard-scaled.

C. Stratified 5-Fold Cross-Validation Splitting

Splits are generated using StratifiedKFold(n_splits=5, shuffle=True, random_state=42), preserving exact cohort prevalence across every partition (WDBC: 37.26% malignant; Heart: 45.87% disease present). Indices are persisted to configs/folds_wdbc.json and configs/folds_heart.json.

IV. QUBIT-CONSTRAINED DIMENSIONALITY REDUCTION

A. Principal Component Analysis (PCAReducer)

PCA projects the feature space into $k \in [4, 8]$ orthogonal eigenvectors capturing maximal sample variance. Principal components are scaled via MinMaxScaler into $[-\pi, \pi]$ to match the periodic boundary conditions of Member B's AngleEmbedding and ZZFeatureMap circuits. For WDBC, k=4 retains 79.24% of variance, while k=8 preserves 92.60%.

B. Mutual Information Feature Selector (MISelector)

To preserve original clinical feature semantics for explainability, we compute Shannon Mutual Information $I(X_j; Y)$ via 3-nearest-neighbors density approximation. The top k ranked features are scaled to $[0, 2\pi]$ for parameterized quantum rotation gates (RX, RY, RZ). Top selected WDBC features: *worst concave points* (0.435 nats), *worst perimeter* (0.404 nats), *mean concave points* (0.401 nats), and *worst radius* (0.369 nats).

V. EXPERIMENTAL BENCHMARKING RESULTS

We evaluated four production-grade classical baseline models across all five stratified validation folds. Table I summarizes performance across accuracy, precision, recall, F1-score, and ROC-AUC.

Dataset	Model Architecture	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	ROC-AUC
WDBC	Classical SVM (RBF)	**97.89 ± 1.2**	**97.71 ± 1.3**	**96.70 ± 1.6**	**97.19 ± 1.5**	0.995 ± 0.005
WDBC	Classical SVM (Linear)	97.54 ± 1.7	97.66 ± 1.3	95.75 ± 1.5	96.73 ± 2.2	**0.996 ± 0.003**
WDBC	PyTorch Deep MLP	97.54 ± 1.2	97.16 ± 1.5	96.24 ± 1.8	96.70 ± 1.5	0.995 ± 0.004
WDBC	XGBoost Classifier	96.13 ± 1.4	96.17 ± 1.6	93.42 ± 2.1	94.79 ± 1.9	0.990 ± 0.007
WDBC	Random Forest	95.43 ± 1.3	95.67 ± 1.6	92.46 ± 2.3	93.85 ± 1.7	0.989 ± 0.006
Heart	Classical SVM (RBF)	**85.48 ± 3.8**	**84.28 ± 3.9**	**83.47 ± 4.1**	**84.07 ± 4.1**	0.908 ± 0.021
Heart	Classical SVM (Linear)	83.83 ± 3.4	83.61 ± 3.5	80.56 ± 4.3	81.43 ± 3.8	**0.911 ± 0.018**
Heart	Random Forest	83.82 ± 3.9	83.33 ± 3.8	80.53 ± 4.3	81.83 ± 4.3	0.906 ± 0.019
Heart	XGBoost Classifier	82.50 ± 2.9	81.98 ± 3.6	79.13 ± 4.4	80.35 ± 3.2	0.902 ± 0.022
Heart	PyTorch Deep MLP	82.18 ± 1.2	80.25 ± 4.0	81.28 ± 4.2	80.52 ± 1.5	0.876 ± 0.025

VI. FOLD INTEGRITY AUDIT & CRYPTOGRAPHIC PARITY

To ensure zero data leakage and exact split parity with Member B's quantum circuits, FoldIntegrityAuditor mathematically validates: (1) partition disjointness, (2) exhaustive coverage, and (3) fold-isolated preprocessing. Persistent fold configurations configs/folds_wdbc.json (SHA-256: 20875b1120fef7381ae4262143bcba161ea87d100067aaae03893c5917804473) and configs/folds_heart.json (SHA-256: fce3882741ea014e76a6cf4ec48ee5f8f8705f157adbb84d5df65d3a95bf2e60) guarantee zero experimental divergence.

VII. EXPLAINABLE AI & CLINICAL INTERPRETABILITY

In clinical decision support systems, model explainability is imperative. Pairing our MISelector with Member C's SHAP audit confirms that high mutual information scores directly reflect recognized physiological pathology: (1) In breast oncology, nuclear concavity and perimeter irregularities indicate cellular pleomorphism and structural invasiveness; (2) In coronary cardiology, exercise-induced ST depression (*oldpeak*) and fluoroscopic major vessel counts (*ca*) correlate directly with severe myocardial ischemia. Our Streamlit controller (app/pages/2_Features.py) allows clinicians to interactively inspect 3D patient embeddings.

VIII. INGESTION SANITIZATION & INFERENCE HARDENING

To ensure safe clinical deployment, DataValidator enforces strict guardrails: (i) detection of missing columns (MissingColumnError), (ii) rejection of malformed or non-numeric types (InvalidDataTypeError), and (iii) enforcement of physiological ranges (OutOfRangeSensorError, e.g. rejecting systolic BP > 260 mm Hg or heart rate > 260 bpm). Both single-patient dictionaries and batch frames are validated with zero-overhead sanitization.

IX. CONCLUSION & ROADMAP

Member A has successfully delivered bulletproof, leak-free data loaders, reproducible 5-fold stratification, four high-performance classical baseline classifiers, dual qubit-reduction modules, an interactive 3D visualization dashboard, and inference hardening guardrails. This solid foundation provides the empirical benchmark against which Member B's quantum kernels and circuits will be evaluated.

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